

E library system

Project Report

Submitted to the Faculty of Computer Technology

For the partial fulfilment of the requirements of

BACHELOR OF COMPUTER APPLICATION

Submitted To:

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BBK DAV COLLAGE FOR WOMEN, LAWRENCE ROAD

AMRITSAR

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DECLARATION

I hereby declare that the work embodied in this project entitled “E-LIBRARY SYSTEM” submitted to the Department of Computer Science, Bbk Dav College For Women, Amritsar, for the partial fulfilment of the degree of Bachelor Computer Applications is entirely based on my own work and not submitted elsewhere for the award of any other degree. All ideas and references have been acknowledged.

Date :10 March,2026

Signature :

Neha

Suhani

CERTIFICATE

Certified that **Ms. Neha** Roll Number **11792313831** and **Ms. Suhani** Roll Number **11792313881** of BCA Sem VI has worked on the project entitled, “**E-LIBRARY SYSTEM**”, for the partial fulfilment of the degree of Bachelor of Computer Applications, Semester 6th during year **2025-2026**. The student will be solely responsible for the authenticity and originality of the work submitted.

Date : 10 March,2026

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Dr. Sushil Kumar
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Neha

Suhani

ABSTRACT

The **E-Library System** is a web-based application developed to automate and digitize the traditional library management system. The main purpose of this project is to provide an efficient and user-friendly platform where users can access books and other learning resources online without any physical limitations.

This system is designed with two main modules: **Admin Module** , **librarian Module** and **User Module**. The admin has full control over the system, including managing books, categories, and user records, log in, search for books, and view available resources. The system ensures proper data management and reduces the chances of data loss and human errors.

The frontend of the project is developed using **HTML, CSS, and JavaScript** to create an interactive and responsive user interface. The backend is implemented using **Node.js**, and **MySQL** is used as the database to store and manage all the data efficiently.

The E-Library System helps save time, reducing manual work, and improving accessibility to educational materials. It provides a modern solution to traditional library systems by enabling users to access information anytime and anywhere. This project is especially useful for students, teachers, and institutions for better resource management and learning experience.

TABLE OF CONTENTS

S no.	Title	Page no.
1.	1. Project Introduction a) Feature b) Objective c) Modules	1 - 4
2.	Software development Life cycle	5
3.	Requirement Analysis	6
4.	Feasibility Report a) Technical Feasibility b) Operational Feasibility c) Economical Feasibility	7
5.	Technologies Used a) Front End b) Back End	8 - 11
6.	Analysis a) Structured Analysis b) Data Flow Diagram c) E-R Diagram	12 - 23
7.	Design Data Design(Database Table Snapshots)	24 - 28
8.	Testing a) Why testing is done b) Testing Principles c) Testing Objectives d) Kinds of Testing	29 - 31
9.	Forms and reports Snapshots	32 - 40
10.	Implementation a) Top Down Implementation b) Botton Up Implementation	41
11.	Maintenance	42
12.	Future Scope	43

E-LIBRARY SYSTEM

1. PROJECT INTRODUCTION

The **E-Library System** is a web-based application designed to automate and manage the operations of a traditional library in a digital environment. In the modern era, managing library records manually is time-consuming, error-prone, and inefficient. This system aims to overcome these limitations by providing a centralized platform where all library-related activities can be performed quickly and efficiently.

The system is developed using modern web technologies and provides a user-friendly interface for three types of users: **Admin, Librarian, and Student**. Each user has specific roles and permissions, ensuring proper management and security of data.

The Admin has full control over the system and manages categories, courses, branches, librarians, and students. The Librarian is responsible for managing books and issuing them to students. The Student can log in to the system and view available books.

This project helps in reducing paperwork, improving data accuracy, and saving time. It also ensures better organization of library resources and provides easy access to information.

Librarians can add, update, and manage books, as well as issue and return them digitally. The system stores all data in a database, making it easy to organize and retrieve information. Overall, an e-library system saves time, reduces paperwork, and provides easy access to learning resources anytime and anywhere.

a) Features

The E-Library System provides a wide range of features that make it efficient and userfriendly. Some of the key features are described below:

1. Multi-User System :The system supports three types of users: Admin, Librarian, and Student. Each user has different access rights and functionalities.

2. Admin Control Panel : The Admin panel allows full control over the system. The admin can:

- Add book categories
- Manage degree types
- Manage courses
- Manage branches
- Manage librarians
- Manage students
- Change password and logout

3. Librarian Dashboard The librarian plays an important role in managing the books. The librarian can:

- View dashboard
- Manage books (add, update, delete)
- Issue books to students
- Change password and logout

4. Student Interface Students can easily interact with the system after login. They can:

- View dashboard
- Check available books
- Change password and logout

5. Book Management System Books can be added, updated, and deleted easily. Each book is stored with complete details such as title, author, ISBN, and category.

6. Issue Book System The librarian can issue books to students and maintain records of issued and returned books.

7. Secure Authentication The system provides login functionality for all users, ensuring that only authorized users can access the system.

b) Objectives

The main objectives of the E-Library System are to improve the efficiency of library operations and provide a better experience for users. The objectives are explained below:

1. Easy Access to Information

- The main objective is to provide users access to books, journals, and study materials anytime and from anywhere. Students do not need to visit the physical library, they can access resources online.

2. Digital Management of Records

- The system stores all data such as student details, book information, and issue/return records in a digital database, making management easier and more organized.

3. Time Saving

- E-Library reduces the time required to search, issue, and return books. Users can quickly find books instead of manually checking shelves.

4. Reduce Paperwork

- All records are maintained electronically, which reduces the use of paper and manual registers, making the system more efficient and eco-friendly.

6. Online Issue and Return System

- Students can issue and return books through the system without physical interaction, and all transactions are recorded automatically.

7. Data Security and Accuracy

- The system ensures that all data is stored securely and reduces chances of errors that usually occur in manual record keeping.

c) Modules

The E-Library System is divided into different modules to ensure proper functionality and management. Each module performs a specific task.

1. Admin Module

The Admin module is the most powerful module in the system. It provides complete control over all operations. The Admin can:

- Add and manage book categories
- Manage degree types
- Manage courses
- Manage branches
- Add and manage librarians
- Add and manage students
- Change password
 - Logout

2. Librarian Module

The Librarian module is responsible for handling book-related operations. The librarian can:

- Access dashboard
- Manage books (add, update, delete)
- Issue books to students
- Maintain issue records
- Change password
- Logout

3. Student Module

The Student module provides access to students to interact with the system. Students can:

- Login to the system
- View dashboard
- Check available books
- Change password
- Logout

2. SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

The **Software Development Life Cycle (SDLC)** is a structured process used for developing high-quality software in a systematic and organized manner. It consists of different phases that guide the development of a software system from initial planning to final maintenance. In the development of the **E-Library System**, The following are the main phases of SDLC used in this project:

1. Planning Phase

In this phase, the idea of developing an E-Library System was identified. The main goal was to create a digital platform to manage library operations such as book management, user handling, and issue records. The scope and objectives of the project were clearly defined.

2. Requirement Analysis Phase

In this phase, all the requirements of the system were collected and analyzed. The needs of different users such as Admin, Librarian, and Student were identified. Functional and nonfunctional requirements were defined to ensure smooth system operation.

3. Design Phase

During the design phase, the overall structure of the system was planned. Database design, system architecture, and user interface layouts were created. Diagrams like Data Flow Diagram (DFD) and Entity Relationship (ER) diagram were also designed to represent the system visually.

4. Implementation Phase

In this phase, the actual coding of the system was done using technologies such as Node.js, Express.js, and MySQL. Different modules like Admin, Librarian, and Student interfaces were developed and integrated.

5. Testing Phase

After implementation, the system was tested to identify and fix errors. Different types of testing such as unit testing, integration testing, and system testing were performed to ensure that the system works correctly and efficiently.

3. REQUIREMENT ANALYSIS

Requirement Analysis is one of the most important phases in the Software Development Life Cycle (SDLC). It involves gathering, analyzing, and defining the needs and expectations of users for developing the system. In the **E-Library System**, requirement analysis helps in understanding what functionalities the system should provide and how it should perform.

The requirements of this system are divided into two main categories: **Functional Requirements** and **Non-Functional Requirements**.

1. Functional Requirements

Functional requirements describe the core functionalities that the system must perform. These are as follows:

- The system must provide separate login interfaces for **Admin, Librarian, and Student**.
- The Admin should be able to manage book categories, degree types, courses, branches, librarians, and students.
- The Librarian should be able to manage books (add, update, delete) and issue books to students.
- The Student should be able to log in and view available books.
- The system must store all data securely in the database.
- The system should allow users to change passwords and logout securely.
- The system should maintain records of issued and returned books.

2. Non-Functional Requirements

Non-functional requirements define the quality and performance aspects of the system. These include:

- **Performance:** The system should respond quickly to user requests.
- **Security:** Only authorized users should access the system using login credentials.
- **Usability:** The interface should be simple and user-friendly.
- **Reliability:** The system should work correctly without errors.
- **Scalability:** The system should be able to handle more users and data in the future.

4. FEASIBILITY REPORT

A feasibility study is conducted to determine whether a project is practical and can be successfully implemented. It evaluates different aspects such as technical resources, operational efficiency, and economic cost. In the **E-Library System**, feasibility analysis ensures that the system can be developed and used effectively without any major issues.

The feasibility study is divided into three main types:

a) Technical Feasibility

Technical feasibility refers to the ability to implement the system using available hardware and software resources. The E-Library System is technically feasible because it is developed using widely used and easily available technologies such as **Node.js**, **Express.js**, and **MySQL**. The system does not require high-end hardware or expensive software. It can run on basic computers with internet connectivity. The development tools like Visual Studio Code and MySQL Workbench are freely available and easy to use.

b) Operational Feasibility

Operational feasibility checks whether the system will work effectively in a real-world environment and whether users can operate it easily. The E-Library System is designed with a simple and user-friendly interface, making it easy for Admin, Librarian, and Student to use. Each user has a clearly defined role, which reduces confusion and improves efficiency. The system automates many tasks such as managing books and issuing them, which reduces manual work and saves time.

c) Economical Feasibility

Economic feasibility evaluates the cost required to develop and maintain the system. The ELibrary System is cost-effective because it uses open-source technologies that are freely available. There is no need for expensive hardware or software. The development cost is minimal, and maintenance can also be done easily without high expenses. By reducing manual work and improving efficiency, the system also helps save time and resources in the long run. Therefore, the project is economically feasible.

Overall, the feasibility study shows that the E-Library System is practical, cost-effective, and easy to implement.

5. TECHNOLOGIES USED

The development of the **E-Library System** is based on a combination of modern web technologies that help in creating an efficient, reliable, and user-friendly application. These technologies are broadly classified into two main categories: **Front-End Technologies** and **Back-End Technologies**. Each technology plays an important role in building different parts of the system and ensures smooth communication between users and the database.

a) Front-End Technologies

Front-end technologies are responsible for designing and developing the user interface of the system. They determine how the application looks and how users interact with it. A good front-end design improves usability and enhances user experience.

1. HTML (Hypertext Markup Language)

HTML is the basic building block of any web application. It is used to create the structure of web pages by defining elements such as headings, paragraphs, forms, buttons, and tables. In the E-Library System, HTML is used to design different pages such as login forms, dashboards, book management pages, and user interfaces. It helps in organizing content in a structured and meaningful way so that users can easily understand and interact with the system.

HTML also supports linking between pages, which allows smooth navigation throughout the application.

Advantage of HTML

- Easy to learn and use
- Used to create structure of web pages
- Platform independent
- Supports multimedia (images, videos, audio)
- Uses tags to define elements (like <p>, <h1>)
- Supports hyperlinks (linking pages)
- Works with CSS and JavaScript
- Free and open standard

2. CSS (Cascading Style Sheets)

CSS is used to style and format the web pages created using HTML. It controls the layout, colors, fonts, spacing, and overall appearance of the system.

In this project, CSS is used to make the interface attractive and visually appealing. It helps in designing clean and professional layouts for different modules such as Admin, Librarian, and Student dashboards.

CSS also supports responsive design, which ensures that the application can be viewed properly on different devices such as desktops, tablets, and mobile phones. This improves accessibility and user experience.

Advantage of HTML

- Used to design and style web pages
- Controls colors, fonts, layout, spacing
- Saves time by applying styles to multiple pages
- Supports responsive design (mobile-friendly)
- Allows animations and transitions
- Improves website appearance
- Separates content (HTML) from design
- Easy to maintain and update

3. JavaScript

JavaScript is a scripting language used to add interactivity to web pages. It allows the system to respond to user actions such as clicking buttons, filling forms, and navigating between pages.

In the E-Library System, JavaScript is used for form validation, which ensures that users enter correct data before submitting forms. It helps in reducing errors and improving data accuracy. JavaScript also enhances user experience by enabling dynamic updates without reloading the page. It plays an important role in making the system more interactive and responsive.

Advantage of JavaScript

- Lightweight and fast scripting language
- Used to make web pages interactive
- Runs on the client-side (browser)
- Easy to learn and use
- Supports event handling (click, hover, etc.)
- Works with HTML and CSS
- Platform independent

4 Bootstrap

Bootstrap is used in an E-Library System to design a responsive and user-friendly interface. It helps in creating attractive web pages with less coding effort. Using Bootstrap, developers can easily design forms for login and registration. It also provides ready-made components like buttons, tables, and navigation bars. These components help in displaying books, students, and issue records clearly.

Bootstrap supports responsive design, so the system works well on mobile, tablet, and desktop. It improves the overall look and feel of the website. Modals in Bootstrap are used for popups like search and alerts. It saves development time by providing pre-designed templates. Overall, Bootstrap makes the E-Library System efficient, modern, and easy to use.

Advantages of Bootstrap

- Easy to use and beginner-friendly
- Saves time with ready-made components
- Responsive design (works on mobile, tablet, desktop)
- Consistent design across all pages
- Provides pre-designed UI elements (buttons, forms, tables)
- Cross-browser compatibility
- Customizable according to needs

b) Back-End Technologies

Back-end technologies handle the server-side operations of the system. They are responsible for processing user requests, managing data, and performing logical operations.

1. Node.js

Node.js is a JavaScript runtime environment used to run JavaScript on the server side. It is built on Chrome's V8 engine and is fast and efficient. Node.js is commonly used to build web applications, APIs, and backend services. It supports asynchronous programming, which helps in handling multiple requests at the same time without slowing down the system. It is widely used in projects like E-Library Systems to manage data, handle user requests, and connect with databases like MySQL.

Node.js is a powerful runtime environment that allows JavaScript to run on the server side. It is widely used for building fast and scalable web applications. In the E-Library System, Node.js is used to handle client requests, process data, and interact with the database. It supports asynchronous programming, which allows multiple users to access the system simultaneously without affecting performance.

Node.js improves the speed and efficiency of the application.

Features of Node.js

- Fast execution using Chrome V8 engine
- Asynchronous and non-blocking (handles multiple requests at once)
- Single-threaded but highly scalable
- Uses JavaScript for both frontend and backend
- Open-source and free to use
- Supports REST API development
- Large ecosystem (NPM packages)
- Real-time data handling (chat apps, live updates)

3. MySQL (Database Management System)

MySQL is used in an E-Library System as a database to store and manage all important data such as student details, book information, and issue/return records. It helps in organizing data in tables and allows fast retrieval using SQL queries. MySQL ensures data security, accuracy, and supports multiple users at the same time. It also helps the system to efficiently handle operations like searching books, issuing, and returning books.

MySQL is a relational database used to store and manage data in the system. It plays a crucial role in the E-Library System. All-important data such as book details, user information, categories, and issue records are stored in MySQL tables. It ensures data consistency, security, and easy retrieval of information. MySQL supports structured queries, which makes it easy to perform operations such as inserting, updating, deleting, and retrieving data.

Features of MySQL

- Open-source and free to use
- Fast and reliable database system
- Supports large amount of data
- Uses Structured Query Language (SQL)
- Easy to use and manage
- Supports multiple users at the same time
- Provides high security (user authentication, access control)

6. SYSTEM ANALYSIS

System Analysis is an important phase in the development of the E-Library System. It involves studying the existing system, identifying problems, and understanding user requirements. The main purpose of system analysis is to design a system that meets user needs efficiently and effectively. The E-Library System is analyzed by dividing it into different components such as structured analysis, data flow diagrams, and entity relationship models.

a) Structured Analysis

Structured analysis is a method used to understand the system in a clear and organized way. It breaks the system into smaller parts called modules so that each part can be studied easily.

In the E-Library System, the system is divided into three main modules:

- Admin Module
- Librarian Module
- Student Module

Each module performs specific functions and works together to form the complete system.

1.Admin Module Analysis

The Admin module is the most important part of the system. It has full control over all operations. The admin manages categories, courses, branches, librarians, and students.

2.Librarian Module Analysis

The Librarian module is responsible for managing books and issuing them to students. It handles operations such as adding new books, updating book details, and maintaining issue records.

3.Student Module Analysis

The Student module allows users to view available books and interact with the system. Students can log in, check book availability, and manage their accounts.

Structured analysis helps in simplifying the system and makes it easier to design and implement.

b) Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows within a system. It shows how data is processed and transferred between different components. A Data Flow Diagram (DFD) is a graphical representation that depicts the flow of data within a system. It's a visual tool used in systems analysis and design to illustrate how data moves between different processes, data stores, and external entities within a system. DFDs are particularly useful for understanding the overall structure and functionality of a system at a high level.

Here are the main components of a DFD:

External Entities:

External entities represent the sources or destinations of data outside the system being modelled. These could be users, other systems, or external data sources with which the system interacts.

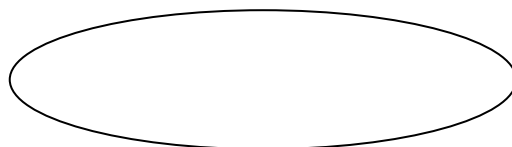
SYMBOL:



Processes:

Processes represent the activities or transformations that occur within the system. Each process takes input data, performs some action on it, and produces output data. Processes are typically labelled with verb phrases that describe the action they perform.

SYMBOL:



Data Flows:

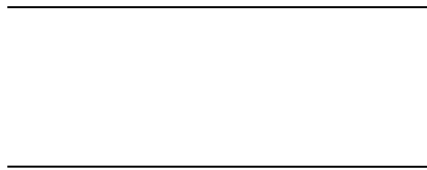
Data flows represent the movement **of** data between processes, data stores, and external entities. They show how data is input to a process, processed, and output to other processes or external entities. Data flows are typically labelled with noun phrases that describe the data being transmitted.

SYMBOL:

**Data Stores:**

Data stores represent repositories where data is stored within the system. These could be databases, files, or other storage mechanisms. Data stores are depicted as rectangles with a name that describes the stored data.

SYMBOL:



ZERO LEVEL DFD:

A **Zero-Level Data Flow Diagram (DFD)**, also known as a **context diagram**, is the highest level of abstraction in a DFD. It provides a broad overview of the system and shows the system as a single process with external entities interacting with it. In this diagram, there are no detailed processes or data stores depicted; rather, it focuses on how the system as a whole communicates with external actors, which can be people, systems, or other entities.



Student/user:

- Secure **login/logout system**
- Password protection and validation
- Request to borrow books
- View issued books list

Librarian

- Add new books image
- Update book details
- Delete books
- Issue books to students

Admin

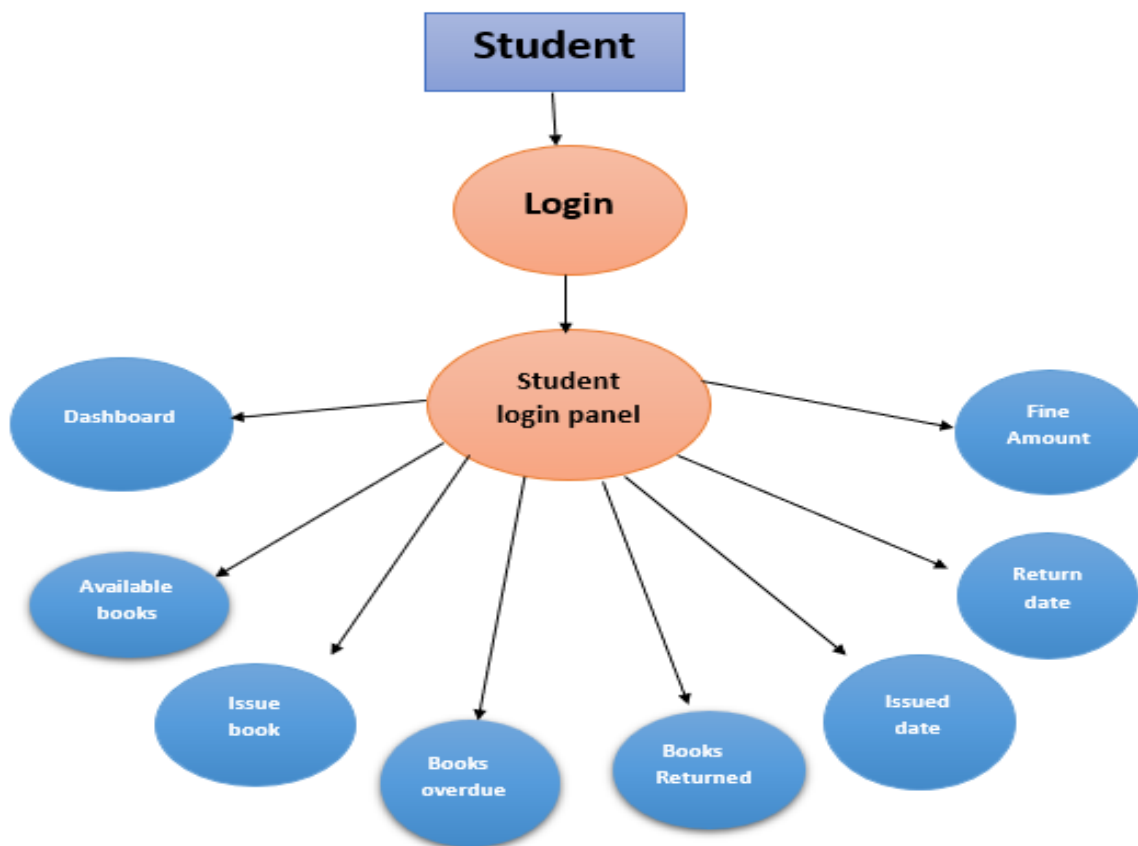
- Manage all users (students & librarians)
- Add / edit / delete books
- Add categories, courses, branches
- View all issued books

1st Level DFD:

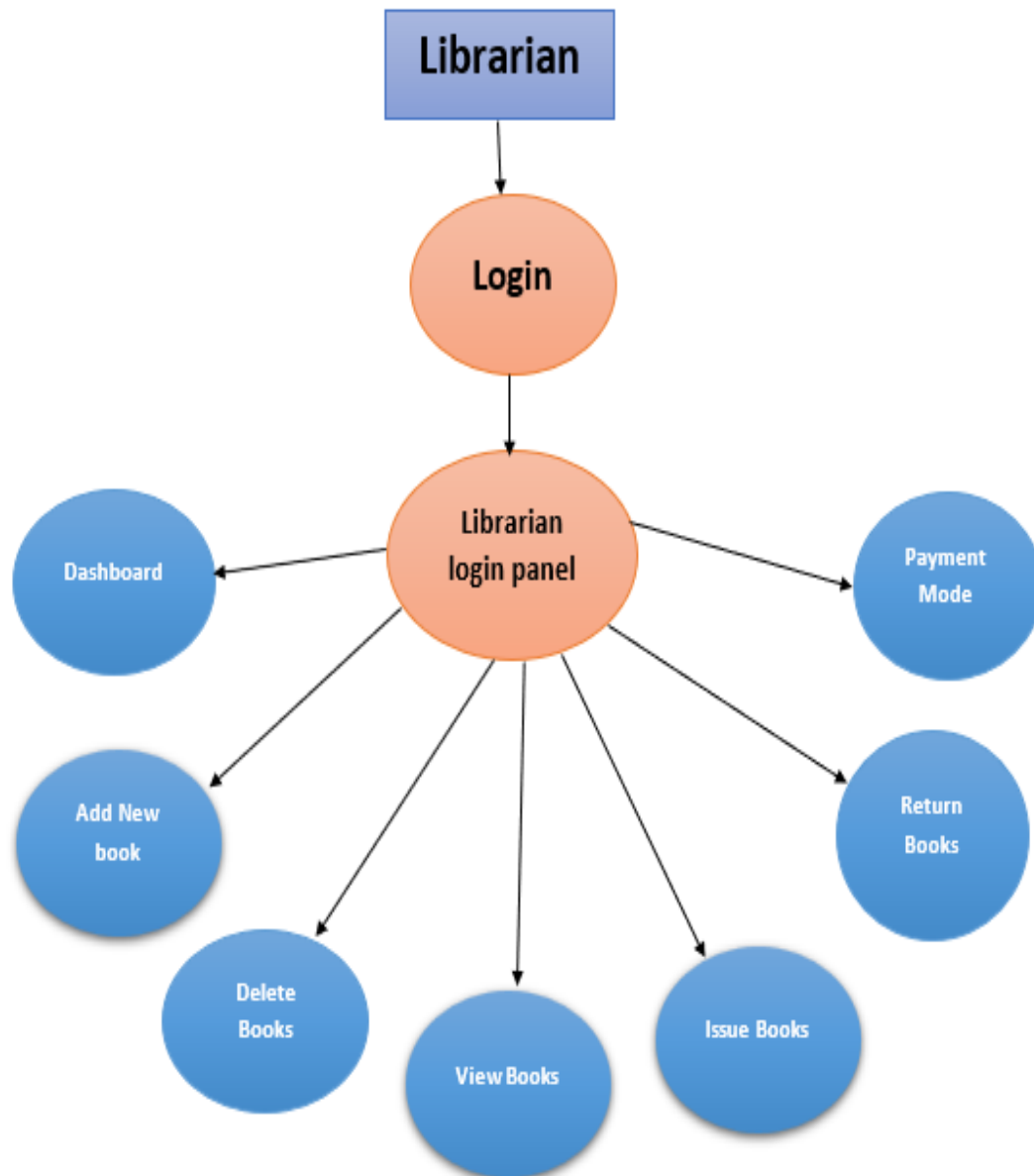
A Level 1 Data Flow Diagram (DFD) is a type of diagram used in systems analysis and design to represent the flow of data within a system at a higher level of abstraction. It provides a more detailed view than the Context Diagram but is still relatively high-level

compared to lower-level DFDs. A Level 1 DFD is created by decomposing the processes of the Context Diagram into more detailed subprocesses. While it provides more detail than the Context Diagram, it still maintains a high level of abstraction, focusing on the major functions and data flows within the system without getting into the intricacies of individual algorithms or data structures.

Student Level DFD:



Librarian Level DFD:



Admin Level DFD:



c) E-R Diagram

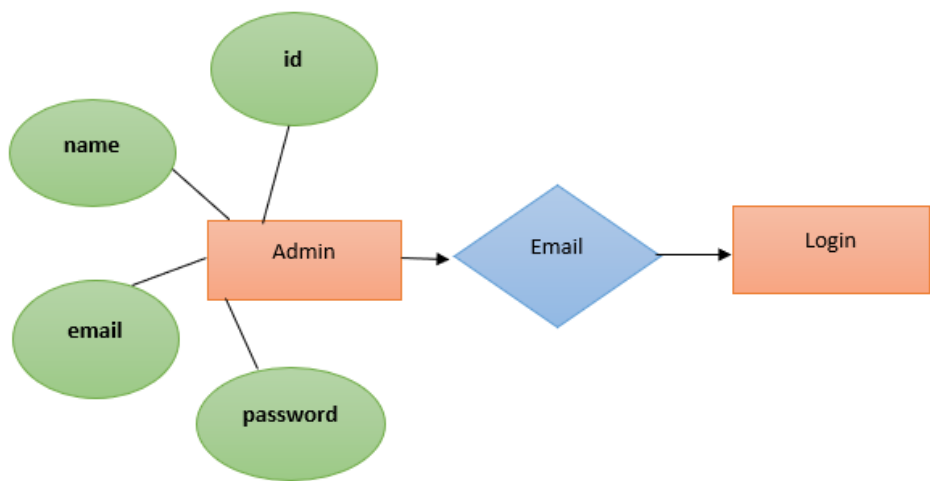
An ER diagram, which stands for Entity-Relationship diagram, is a visual representation used to model the data or information requirements and the relationships between entities within a system. ER diagrams are commonly used in database design to illustrate the logical structure of a database.

The main components of an ER diagram are:

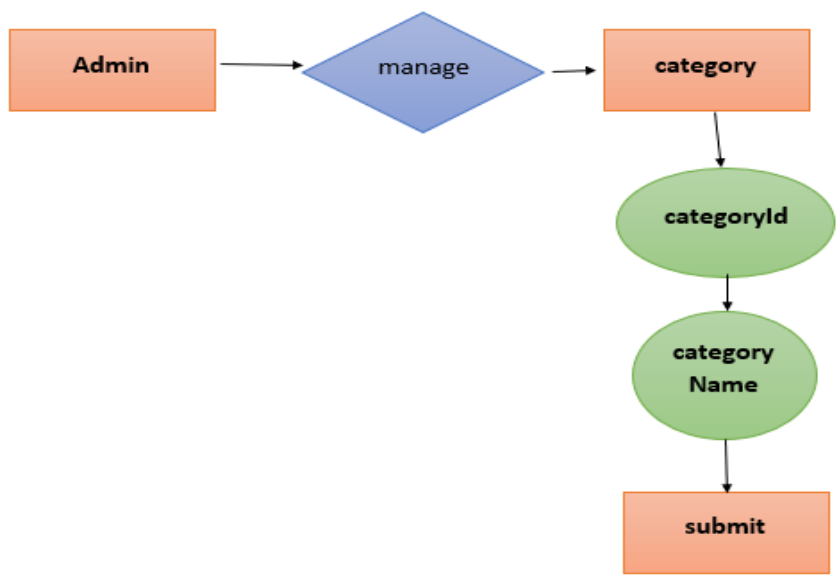
- **Entities:** These are the objects or concepts about which data is stored in the database. Entities are typically represented by rectangles in the diagram.
- **Attributes:** Attributes are the properties or characteristics of entities. They describe the data that can be stored for each entity. Attributes are usually depicted as ovals connected to their respective entities.
- **Relationships:** Relationships describe how entities are related to each other within the system. They represent the associations between entities. Relationships are typically depicted by lines connecting the related entities, and they may have a name and cardinality (such as one-to-one, one-to-many, or many-to-many).
- **Keys:** Keys are attributes (or combination of attributes) that uniquely identify each entity within the database. They are crucial for maintaining data integrity and establishing relationships between entities.

Following are the ER Diagrams:

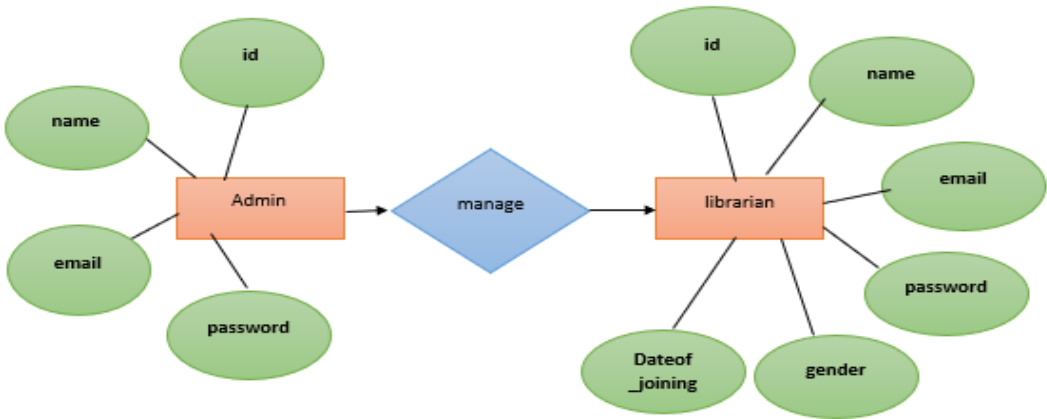
Admin Login



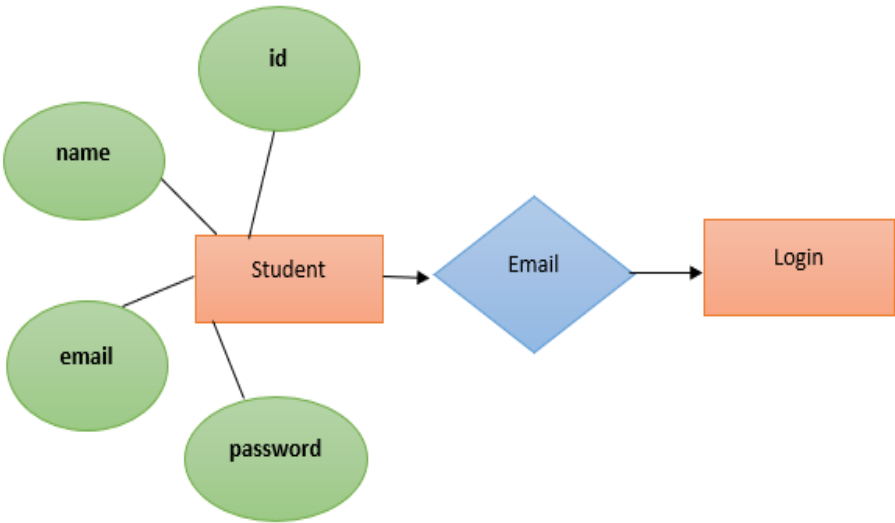
Category



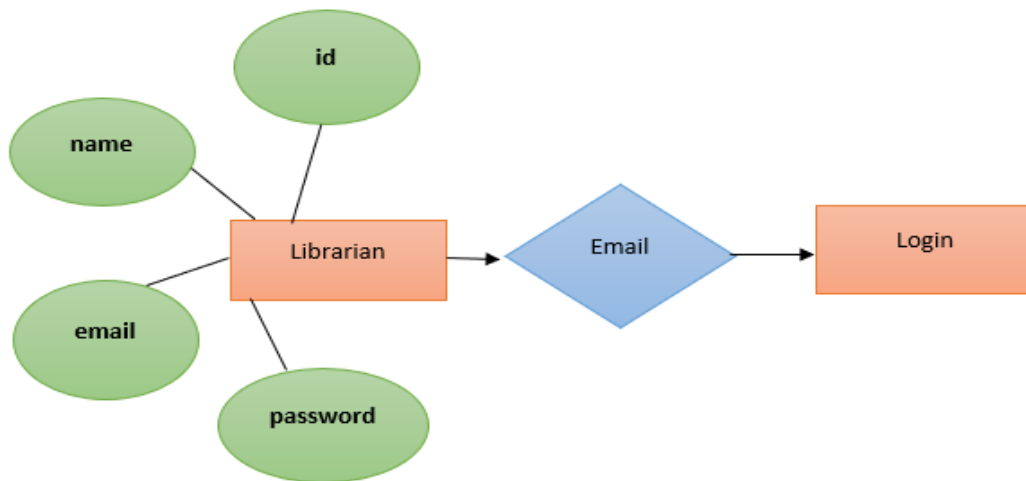
Admin manage Librarian



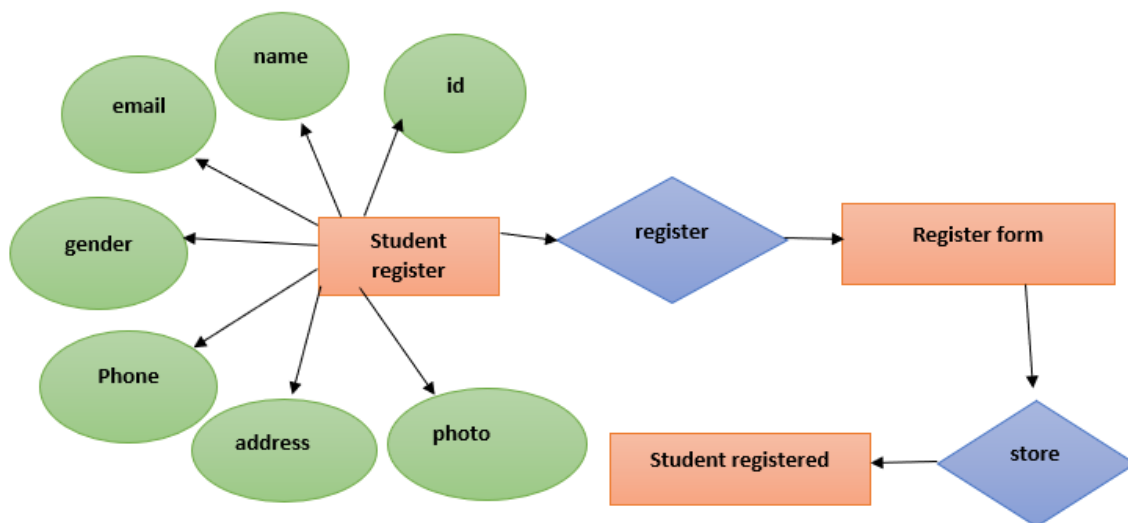
Student Login



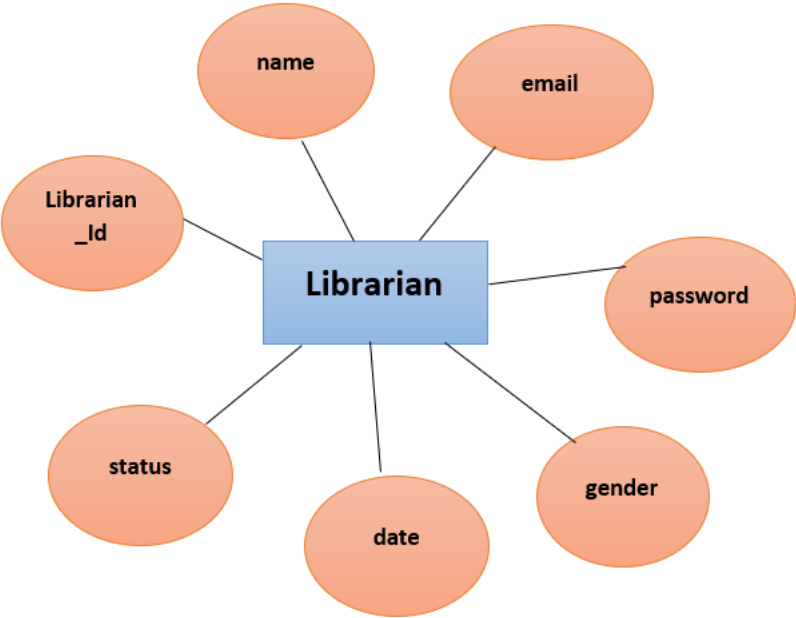
Librarian Login



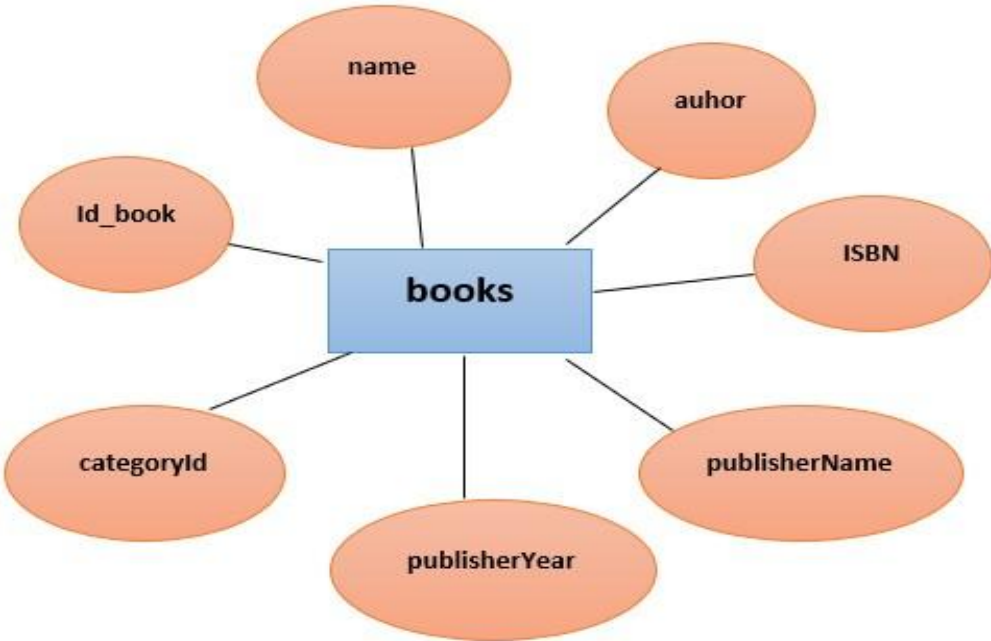
Student Register



Librarian



Books



7. SYSTEM DESIGN

System Design is an important phase in software development where the overall structure of the system is created. It defines how the system will work, how data will be stored, and how different components will interact with each other.

The design of the **E-Library System** focuses on creating a well-structured, efficient, and scalable system. It ensures that all modules work properly and data is stored in an organized manner.

a) Data Design

Data design refers to the process of organizing and structuring data in a way that it can be stored, managed, and retrieved efficiently. In the E-Library System, data is stored using a relational database (MySQL).

The database is divided into multiple tables, each representing a specific entity. These tables are connected using relationships to ensure data integrity.

1.Admin Table

S.No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	name	varchar (255)	Not Null
3.	email	varchar (255)	Null Allowed
4.	password	varchar (255)	Null Allowed
5.	status	varchar (255)	Default 'Active'
6.	createdAt	datetime	Not Null
7.	updatedAt	datetime	Not Null

2. Degree Table

S.No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	name	varchar (255)	Not Null
3.	createdAt	datetime	Not Null
4.	updatedAt	datetime	Not Null

3. Course Table

S. No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	name	varchar (255)	Not Null
3.	degreeId	int	Foreign Key, Not Null
4.	createdAt	datetime	Not Null
5.	updatedAt	datetime	Not Null

4. Branch Table

S. No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	name	varchar (255)	Not Null
3.	courseId	int	Foreign Key, Not Null
4.	createdAt	datetime	Not Null
5.	updatedAt	datetime	Not Null

5. Librarian Table

S.No.	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	name	varchar (255)	Not Null
3.	email	varchar (255)	Not Null
4.	password	varchar (255)	Null Allowed
5.	gender	varchar (255)	Null Allowed
6.	date	datetime	Null Allowed
7.	status	varchar (255)	Default 'Active'
8.	createAt	datetime	Not Null
9.	updateAt	datetime	Not Null

6. Student Table

S.No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	name	varchar (255)	Not Null
3.	email	varchar (255)	Null Allowed
4.	password	varchar (255)	Null Allowed
5.	gender	varchar (255)	Null Allowed
6.	mobile	varchar (255)	Null Allowed
7.	address	varchar (255)	Null Allowed
8.	city	varchar (255)	Null Allowed
9.	state	varchar (255)	Null Allowed
10.	zip	int	Null Allowed
11.	photo	varchar (255)	Null Allowed
12.	branchId	int	Foreign key , Not Null
13.	status	varchar (255)	Default 'Active', Not Null
14.	createdAt	datetime	Not Null
15.	updatedAt	datetime	Not Null

7. Books Table

S.No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	title	varchar (255)	Not Null
3.	author	varchar (255)	Not Null
4.	status	varchar (255)	Default 'Available', Null Allowed
5.	isbn	varchar (255)	Not Null
6.	publisherName	varchar (255)	Not Null
7.	publishYear	Varchar (255)	Not Null
8.	photo	varchar (255)	Null Allowed
9.	categoryId	Int	Foreign Key, Not Null
10.	createdAt	datetime	Not Null
11.	updatedAt	datetime	Not Null

8.Category Table

S.No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	name	varchar (255)	Not Null
3.	createdAt	datetime	Not Null
4.	updatedAt	datetime	Not Null

9. Issue Table

S.No	Attributes	Datatypes	Constraint
1.	id	Int AI PK	Not Null
2.	userId	int	Foreign key, Not Null
3.	bookId	int	Foreign key,Not Null
4.	issuedAt	datetime	Not Null
5.	dueDate	datetime	Not Null
6.	returnedAt	datetime	Null Allowed
7.	status	enum('Issued','Returned','Overdue','Lost')	Default 'Issued', Null Allowed
8.	fineAmount	decimal(10,2)	Default 0.00, Null Allowed
9.	finePaid	tinyint(1)	Default 0, Null Allowed
10.	paymentMode	enum('Cash','UPI','Card','Online')	Null Allowed
11.	transactionId	varchar (255)	Null Allowed
12.	paidAt	datetime	Null Allowed
13.	createdAt	datetime	Not Null
14.	updatedAt	datetime	Not Null

8. TESTING

Testing is a very important phase in software development. It ensures that the system works correctly, efficiently, and meets the user requirements. In the E-Library System, testing is performed to identify errors, fix bugs, and improve the overall performance of the application. Proper testing helps in delivering a reliable and user-friendly system.

(a) Why Testing is Done

Testing is done to ensure that the developed software is free from errors and functions according to the requirements. In an E-Library System, users depend on the system to search, issue, and manage books. If the system fails or gives incorrect results, it can create problems for both students and librarians.

The main reasons for performing testing are:

- **Error Detection:** Testing helps in identifying bugs and errors in the system before it is used by real users.
- **Quality Assurance:** It ensures that the software meets quality standards and performs efficiently.
- **User Satisfaction:** A well-tested system provides a smooth and reliable user experience.
- **Security Checking:** Testing helps in identifying security issues such as unauthorized access.
- **Performance Improvement:** It ensures that the system works properly even under heavy usage.

In short, testing is essential to make the system reliable, secure, and efficient.

(b) Testing Principles

There are some basic principles that should be followed during the testing process:

1. **Testing Shows Presence of Defects:** Testing can show that defects exist, but it cannot prove that there are no defects in the system.
2. **Exhaustive Testing is Not Possible:** It is not possible to test every input and condition, so testing is done on selected cases.
3. **Early Testing:** Testing should start as early as possible in the development process to detect issues sooner.

4. **Defect Clustering:** Most defects are usually found in a small number of modules or areas of the system.
5. **Pesticide Paradox:** If the same test cases are used repeatedly, they may stop finding new bugs. So, test cases should be updated regularly.
6. **Testing is Context Dependent:** Different types of applications require different testing approaches.
7. **Absence of Errors is a Fallacy:** Even if no errors are found, the system is not useful unless it meets user requirements.

These principles guide testers in performing effective and efficient testing.

(c) Testing Objectives

The main objectives of testing are to ensure that the system works correctly and meets user expectations. In the E-Library System, the objectives include:

- **Verification of Functionality:** To check whether all features like login, book search, issue, and return are working properly.
- **Validation of Requirements:** To ensure that the system meets all the specified requirements.
- **Error Identification:** To find and fix bugs before deployment.
- **Performance Testing:** To check how the system behaves under different conditions.
- **Security Testing:** To ensure that user data is safe and unauthorized access is prevented.
- **Usability Testing:** To make sure that the system is easy to use for both students and administrators.
- **Reliability Testing:** To ensure that the system works consistently without crashing.

The main goal is to deliver a high-quality and user-friendly system.

(d) Kinds of Testing

There are different types of testing used in the E-Library System:

1. Unit Testing

In unit testing, individual components or modules are tested separately. For example, testing the login function or book search feature independently.

2. Integration Testing

In this testing, different modules are combined and tested together. For example, checking whether the login module works properly with the database.

3. System Testing

The complete system is tested as a whole to ensure that all components work together correctly.

4. Functional Testing

It checks whether all functions of the system are working as expected. For example, adding books, issuing books, and returning books.

5. Performance Testing

This testing checks the speed and performance of the system under different conditions. **6.**

Security Testing

It ensures that the system is protected from unauthorized access and data breaches.

7. User Acceptance Testing (UAT)

This testing is done by users to check whether the system meets their requirements and expectations.

8. Regression Testing

It ensures that new changes or updates do not affect the existing functionality of the system.

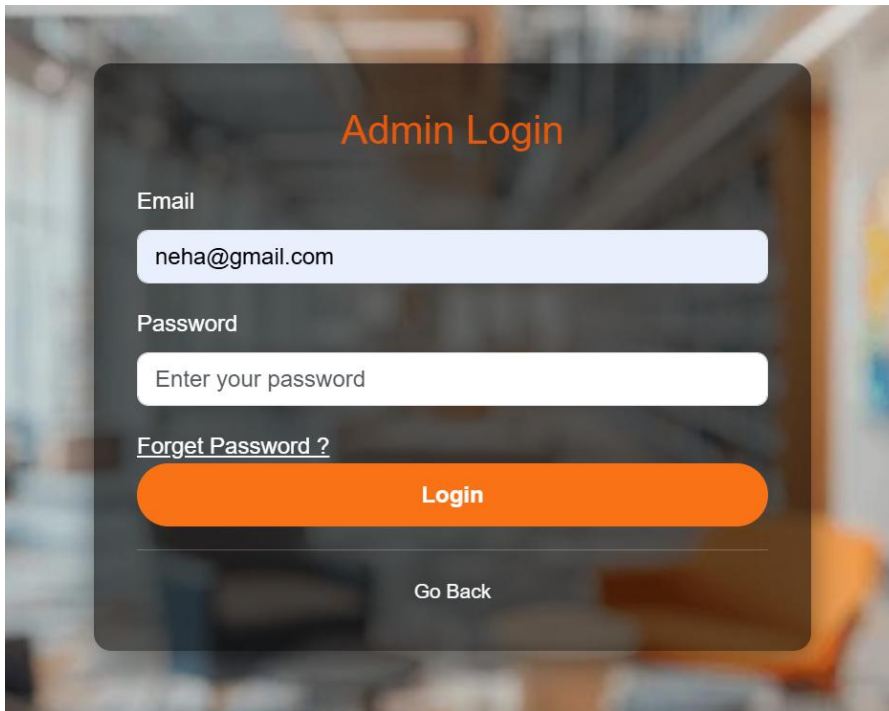
9. FORMS AND REPORTS SNAPSHOTS

The E-Library System consists of various forms and reports that help users and administrators to manage library operations efficiently. Forms are used to input and manage data, while reports are used to display and analyze information in a structured format.

Below are the important forms and reports used in the system:

9.1 Forms in E-Library System

1. Admin Login Form

A screenshot of the Admin Login form. The form is a dark gray rounded rectangle centered on a blurred background of a library. At the top, the title "Admin Login" is written in orange. Below it, there are two input fields: "Email" with the text "neha@gmail.com" and "Password" with the placeholder "Enter your password". To the left of the password field is a link "Forget Password ?" in white. Below the input fields is a large orange "Login" button. At the bottom of the form is a "Go Back" link in white.

Admin Login

Email
neha@gmail.com

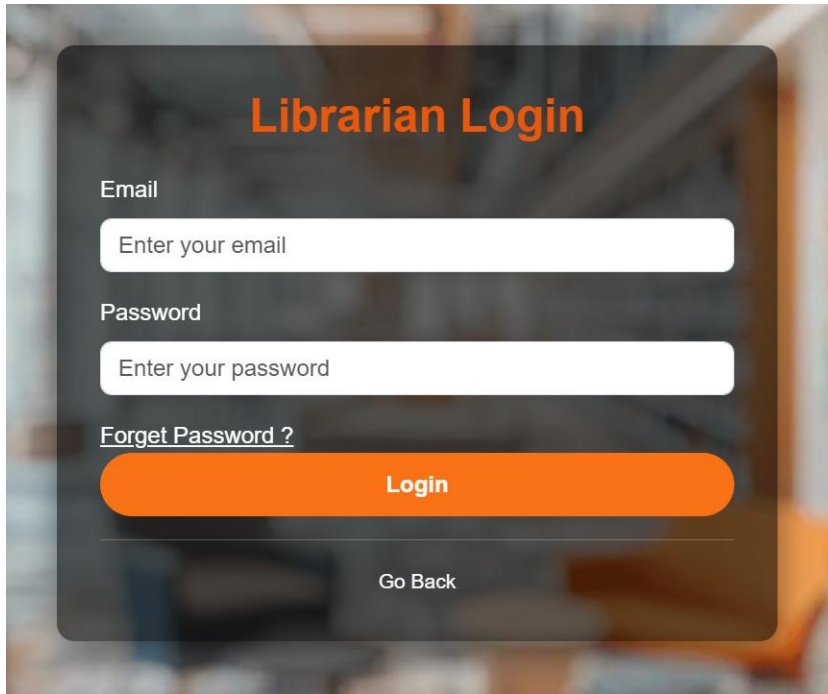
Password
Enter your password

[Forget Password ?](#)

Login

[Go Back](#)

2. Librarian Login Form

A screenshot of a Librarian Login form. The form is a dark gray rounded rectangle centered on a blurred background of a library. At the top, the title "Librarian Login" is written in orange. Below it are two white input fields: "Email" with the placeholder "Enter your email" and "Password" with the placeholder "Enter your password". A link "[Forget Password ?](#)" is positioned below the password field. A large orange "Login" button is centered below the fields, and a "Go Back" link is at the bottom.

Librarian Login

Email

Enter your email

Password

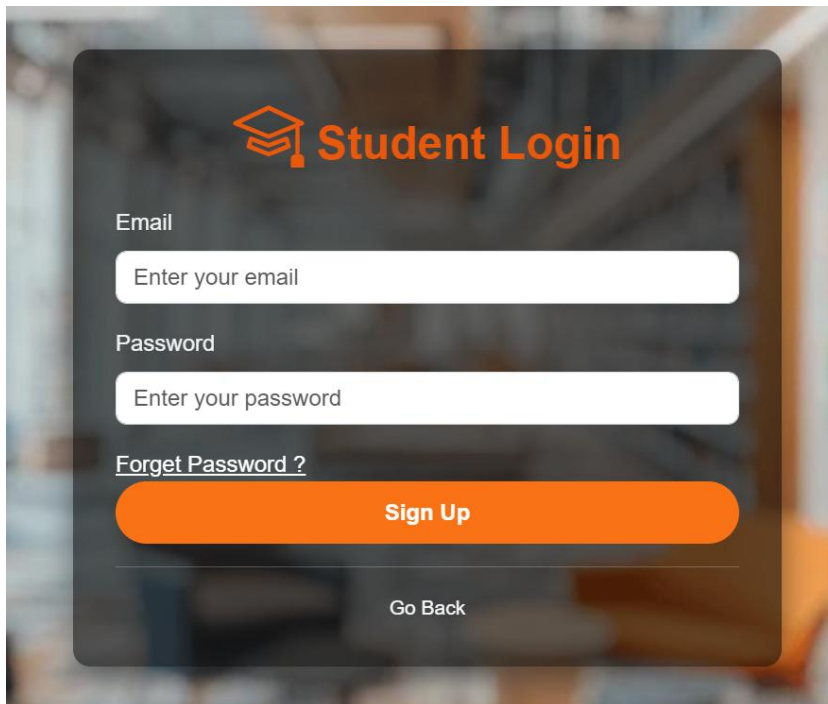
Enter your password

[Forget Password ?](#)

Login

[Go Back](#)

3. Student Login Form

A screenshot of a Student Login form. The form is a dark gray rounded rectangle centered on a blurred background of a library. At the top, there is an orange graduation cap icon followed by the title "Student Login" in orange. Below it are two white input fields: "Email" with the placeholder "Enter your email" and "Password" with the placeholder "Enter your password". A link "[Forget Password ?](#)" is positioned below the password field. A large orange "Sign Up" button is centered below the fields, and a "Go Back" link is at the bottom.

 **Student Login**

Email

Enter your email

Password

Enter your password

[Forget Password ?](#)

Sign Up

[Go Back](#)

4. Add Librarian Form

Add New Librarian

Name	Email	Select Gender
Enter your name	Enter your email	Choose... ▼
Date Of joining		
dd-mm-yyyy 📅		
Add Librarian		

5. Manage Student Form

Add New Student

Select Degree-Type	Select Course	Select Branch
Select category ▼	▼	▼
Name	Email	Gender
Enter your name	Enter your email	Please Select Gender ▼
Mobile	Address	City
Enter your number	Enter your address	Enter your city
State	Zip	Photo
Enter your state	Enter your zip	Choose File No file chosen
Register student		

6. Add new Book

Add New Book

Book Title	Author Name(s)
Book Title	Author Name(s)
ISBN Number (unique book identifier)	Category / Genre
ISBN Number	Select category
Publisher Name	Publication Year
Publisher Name	Publisher Year
Cover Photo (jpg/png)	
Choose File No file chosen	
Save	

7. Issue Books Form

Issue Books - Student Detail

Select Degree-Type	Select Course-Type	Select Branch
Select Degree	Select Course	Select Branch
Select Student	Select Book Category	Select Book
Select Student	Select Category	Select Book
Issue Date	Return Date	
Issue Book		

9.2 Reports in E-Library System

1. Category Report

List of Category

Id	Name	Delete
12	Artificial intelligence	Delete
11	General Knowledge	Delete
10	Children	Delete
9	Motivational	Delete
8	Novel	Delete
7	Literature	Delete
6	Biography	Delete
5	Software Engineering	Delete
4	Operating system	Delete
3	History	Delete
2	Programming	Delete
1	Computer Scinece	Delete

2. Books Report

List of Books

Category / Genre	Book Title	Author Name(s)	ISBN Number	Publisher Name	Publication Year	Photo	Status	
Artificial Intelligence	Artificial Intelligence: A Modern Approach	Stuart Russell	9780134610993	Pearson	2020		Available	Delete
General Knowledge	Manorama Yearbook	K.M. Mathew	9788194833666	Malayala Manorama	2022		Issued	Delete
Motivational	Think and Grow Rich	Napoleon Hill	9781585424337	TarcherPerigee	2005		Available	Delete
Children	Harry Potter and the Sorcerer's Stone	J.K. Rowling	9780590353427	Scholastic	1998		Available	Delete
Novel	The Alchemist	Paulo Coelho	9780062315007	HarperCollins	2014		Available	Delete
Literature	Pride and Prejudice	Jane Austen	9780141439518	Penguin Classics	2003		Available	Delete
Operating system	Operating System Concept	Abraham Silberschatz	9780073523323	wiley	2018		Available	Delete
Software Engineering	Software Engineering: A Practitioner's Approach	Roger S. Pressman	9780078022128	McGraw-Hill	2014		Issued	Delete
Biography	wings of Fire	A.P.J.Abdul Kalam	9788173711466	Universities Press	1999		Issued	Delete
History	The Discovery of India	Jawaharlal Nehru	9780143031031	Signat Press	1946		Available	Delete
Programming	c++	Anshuman gupta	9781337102087	Cengage learning	2017		Available	Delete
Computer Scinece	JavaScript: The Good Parts	Douglas Crockford	9780596517748	O'Reilly Media	2008		Available	Delete
Computer Scinece	Clean Code	Robert C. Martin	9780132350884	Pearson	2008		Available	Delete

3. Issued Books Report

Issued Books to Students									
Id	Student Name	Book Name	Issued Date	Return Date	Fine	Payment Mode	Transaction Id/Remarks	Status	Action
19	Amit	Manorama Yearbook	2026-03-29	2026-04-02	30.00	-	-	Overdue	Return Book
18	Prabh	Software Engineering: A Practitioner's Approach	2026-03-31	2026-04-02	30.00	-	-	Overdue	Return Book
17	Amit	wings of Fire	2026-04-03	2026-04-06	0.00	-	-	Issued	Return Book
16	Neha	c++	2026-03-28	2026-04-01	20.00	Cash	12345	Returned	
15	Amit	Clean Code	2026-04-02	2026-04-05	0.00	-	-	Returned	
14	Neha	c++	2026-04-01	2026-04-04	0.00	-	-	Returned	
13	Suhani	JavaScript: The Good Parts	2026-03-28	2026-03-31	20.00	UPI	12345	Returned	

4. Student Report

List of Student										
Branch	Name	Email	Gender	Mobile	Address	City	State	Zip	Photo	Status
BCA	Suhani	suhani30957@gmail.com	Female	1234567890	mali gali	amritsar	punjab	143001		Active
BCA	Neha	nehanehaasr1812@gmail.com	Female	1234554321	mohkapura	Amritsar	Punjab	143001		Active
BCA	Prabh	prabhjyotkaurp477@gmail.com	Female	5435476234	Majitha road	Amritsar	Punjab	143001		Active
IT	Amit	aak524025@gmail.com	Male	2345654323	Krishna Nagar	amritsar	Punjab	143001		Active
Marketing	Ankit	kumarankit73970@gmail.com	Male	5432456789	Krishna Nagar	Amritsar	Punjab	143001		Active
IT	Neha	nehaasr1812@gmail.com	Female	3456734567	Mohkapura	Amritsar	Punjab	143001		Active

5. Librarian Report

List of Librarian

Id	Name	Email	Gender	Date Of Joining	Status	
6	Neha	nehanehaasr1812@gmail.com	Female	02-04-2026	Active	Deactivate
5	Suhani	suhani30957@gmail.com	Female	02-04-2026	Active	Deactivate

6. Course Report

List of Courses

Degree Name	Course Name	
UG	B.Com	Delete
UG	B.Sc	Delete
UG	B.Voc	Delete
UG	BCA	Delete
UG	B.Tech	Delete
PG	MCA	Delete

7. Branch Report

List of Branch

Course Name	Branch Name	
B.Com	Marketing	Delete
B.Voc	IT	Delete
B.Sc	IT	Delete
BCA	BCA	Delete
MCA	MCA	Delete
B.Tech	CSE	Delete

8. Degree Report

List of Degree Type

Name	Delete
Diploma	Delete
UG	Delete
PG	Delete

10. IMPLEMENTATION

Implementation is the phase in which the actual system is developed and executed based on the design specifications. In this phase, all the modules of the **E-Library System** are coded, integrated, and made ready for use. It involves converting the system design into a working application using programming languages and technologies such as Node.js, Express.js, and MySQL.

Implementation ensures that all components of the system work together properly and provide the expected output. It also involves testing and debugging to remove errors and improve performance.

a) Top-Down Implementation

Top-down implementation is an approach in which the system is developed starting from the main modules and then moving towards smaller sub-modules. In this method, the overall structure of the system is created first, and then detailed functionalities are added step by step. In the E-Library System, the top-down approach is used by first developing the main modules such as Admin, Librarian, and Student interfaces. After that, sub-modules like book management, category management, and issue system are implemented.

This approach helps in understanding the overall structure of the system and ensures that all major functionalities are planned before implementation. It also makes it easier to manage and organize the system efficiently.

b) Bottom-Up Implementation

Bottom-up implementation is an approach in which the system is developed starting from the smaller modules and then integrating them to form the complete system. In this method, individual components or functions are created first and then combined to build higher-level modules.

In the E-Library System, bottom-up implementation is used by first developing basic functionalities such as database connectivity, user authentication, and individual operations like adding books, updating records, and managing data. After that, these small components are integrated to form complete modules like Admin, Librarian, and Student panels.

This approach ensures that each small part of the system is tested and working correctly before integration. It improves reliability and reduces errors during development. It also makes debugging easier because issues can be identified at the component level.

11. MAINTENANCE

Maintenance is the final phase of the Software Development Life Cycle (SDLC) that takes place after the successful implementation of the system. It involves updating, improving, and modifying the system to ensure its smooth functioning over time. In the **E-Library System**, maintenance plays an important role in keeping the system efficient, secure, and up-to-date.

Maintenance is required because software systems may face issues such as bugs, performance problems, or changes in user requirements after deployment. These issues need to be identified and resolved to maintain system quality.

There are different types of maintenance performed in the system:

1. **Corrective Maintenance** This type of maintenance is used to fix errors or bugs found in the system after deployment. For example, if there is an issue in login functionality or book issuing process, it is corrected under this maintenance.
2. **Adaptive Maintenance** Adaptive maintenance is done when changes are required due to new technologies or environment changes. For example, updating the system to work with new database versions or browsers.
3. **Perfective Maintenance** This maintenance focuses on improving the performance and functionality of the system. New features such as advanced search options or improved user interface can be added.
4. **Preventive Maintenance** Preventive maintenance is performed to avoid future problems. It includes regular system checks, updates, and improvements to ensure stability.

Maintenance ensures that the E-Library System continues to work efficiently and meets user needs. It also helps in improving system performance, enhancing security, and extending the life of the software.

12. FUTURE SCOPE

Future scope refers to the possible improvements and enhancements that can be made to the system in the future. The **E-Library System** has a wide scope for further development and can be upgraded with advanced features to make it more efficient and user-friendly.

One of the major improvements can be the development of a **mobile application** so that users can access the system from their smartphones anytime and anywhere. This will increase accessibility and convenience for students.

Another enhancement can be the addition of an **online book reservation system**, which allows students to reserve books in advance. This will help in better management of book availability.

The system can also include **email and SMS notifications** to inform users about book issue, return dates, and overdue reminders. This will improve communication between the system and users.

Integration of a **barcode system** can make book issuing and returning faster and more accurate.

It will reduce manual work and improve efficiency.

The system can also be enhanced by adding **advanced search features** and **filter options** to help users find books easily.

In the future, **AI-based recommendation systems** can be added to suggest books to users based on their interests and previous activities.

Overall, the E-Library System has great potential for future development. With continuous improvements and new features, it can become a more powerful and efficient system.